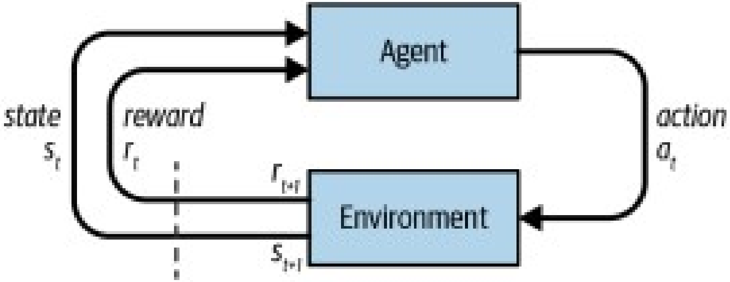

Figure: Reinforcement learning diagram

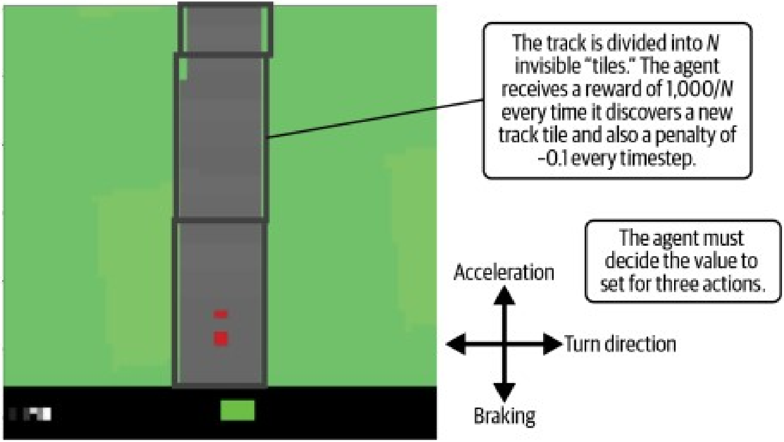

Figure: A graphical representation of one game state in the CarRacing environment

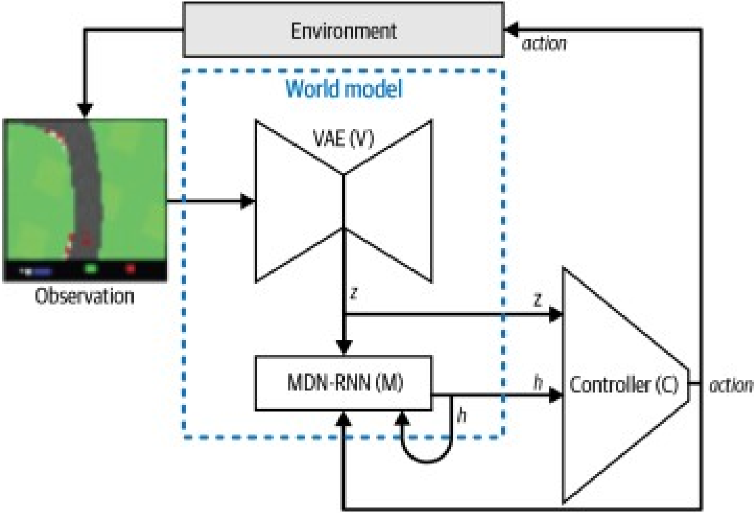

Figure: World model architecture diagram

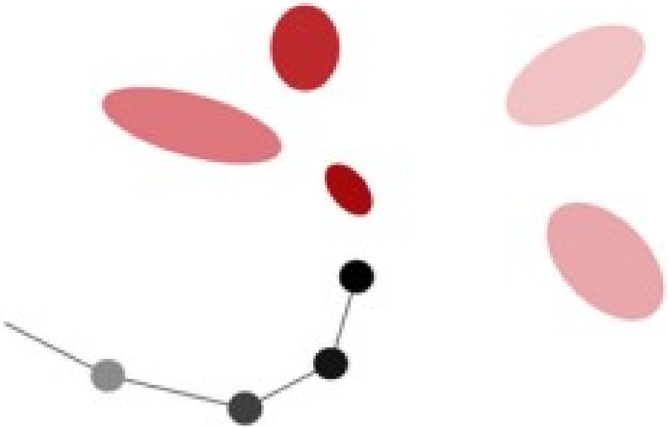

Figure: MDN for handwriting generation

Figure: Frames 40 to 59 of one episode

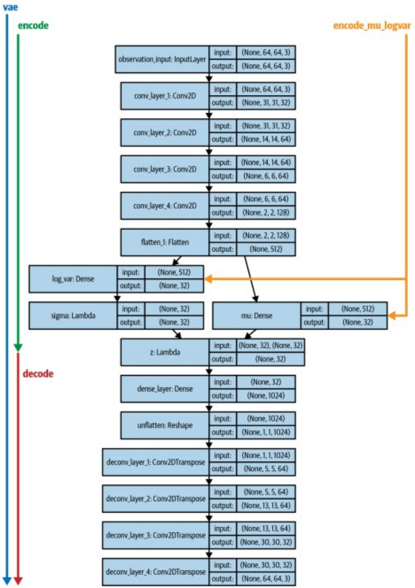

Figure: The VAE architecture from the "World Models" paper

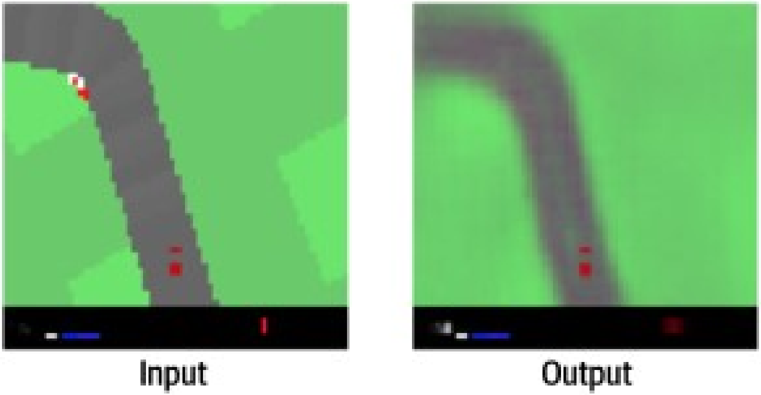

Figure: The input and output from the VAE model

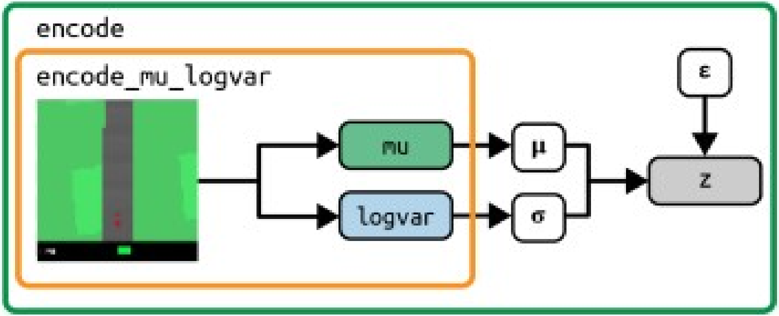

Figure: The output from the encoder models

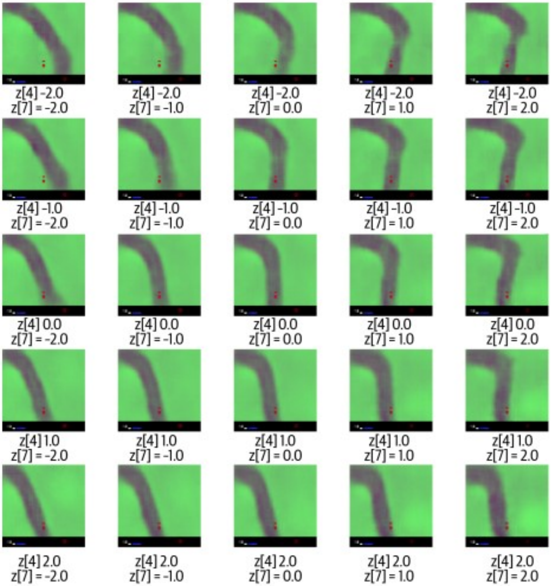

Figure: A linear interpolation of two dimensions of z

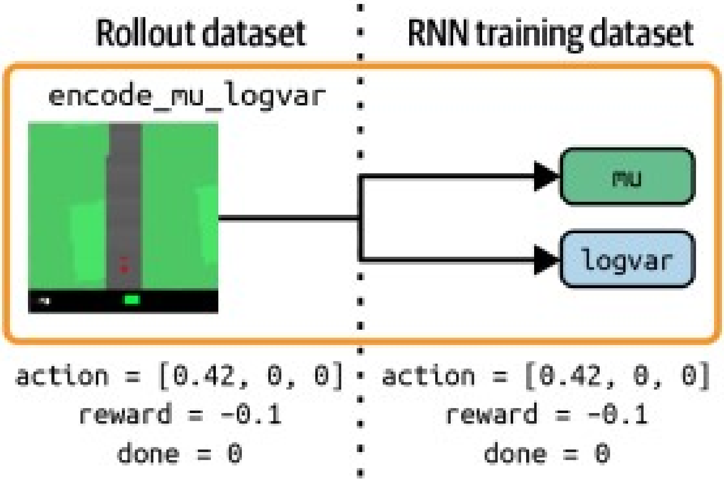

Figure: Creating the MDN-RNN training dataset

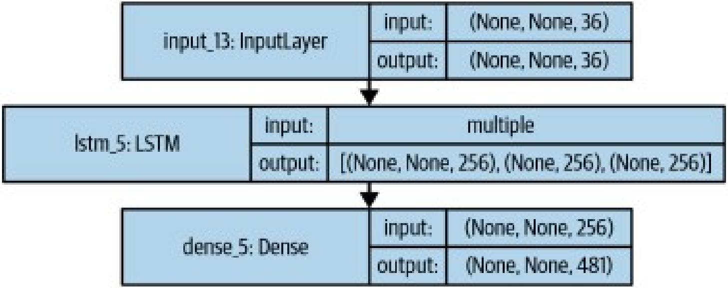

Figure: The MDN-RNN architecture

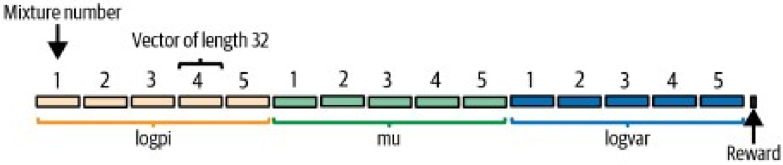

Figure: The output from the mixture density network

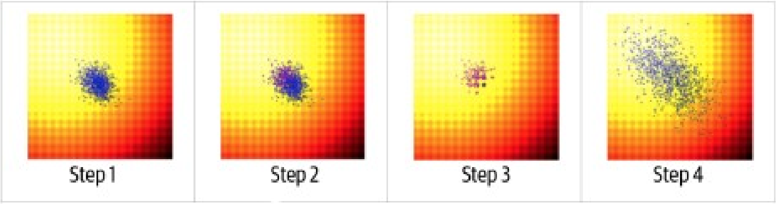

Figure: One update step from the CMA-ES algorithm

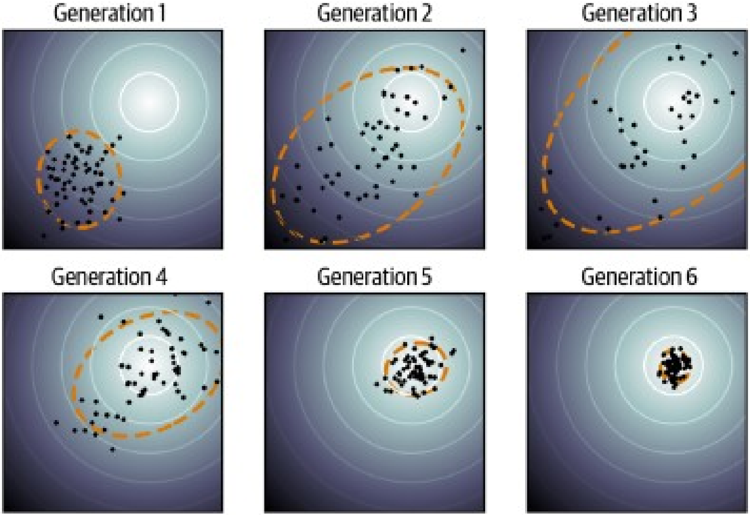

Figure: CMA-ES

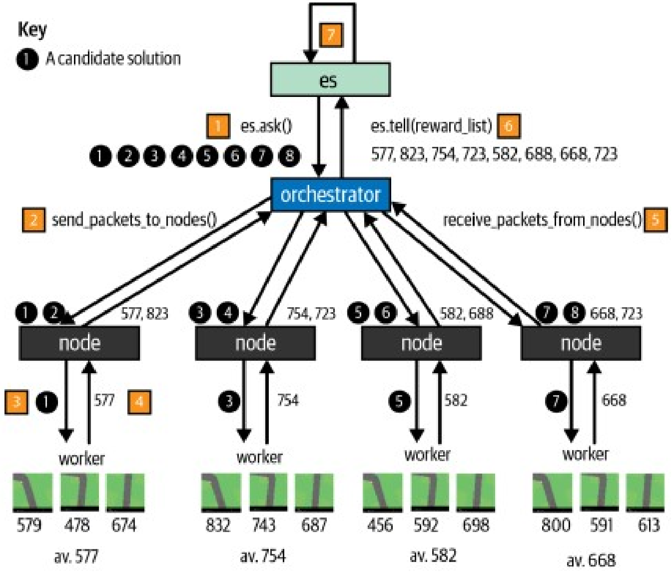

Figure: Parallelizing CMA-ES—here there is a population size of eight and four nodes (so t = 2, the number of trials that each node is responsible for)

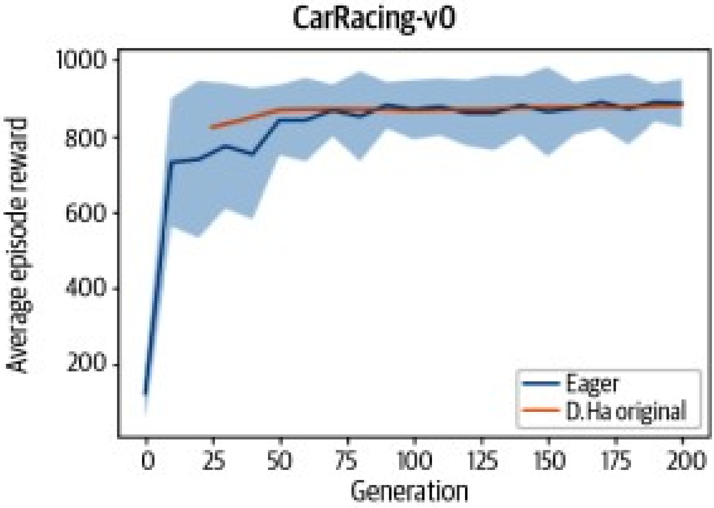

Figure: Average episode reward of the controller training process, by generation

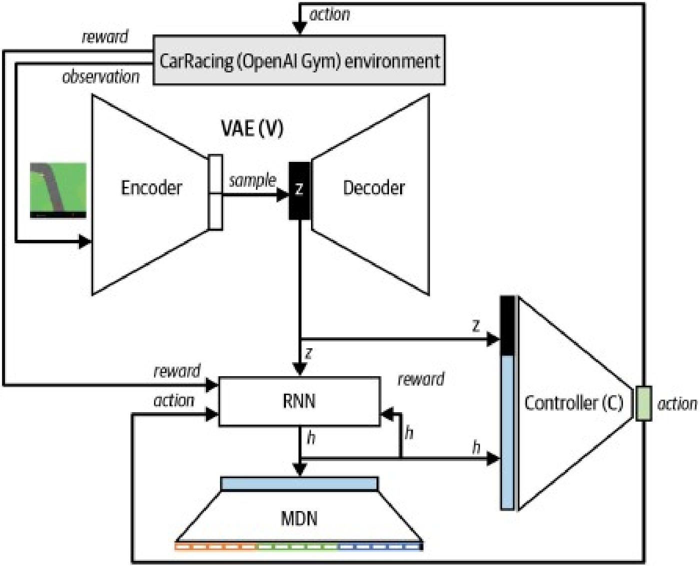

Figure: Training the controller in the Gym environment

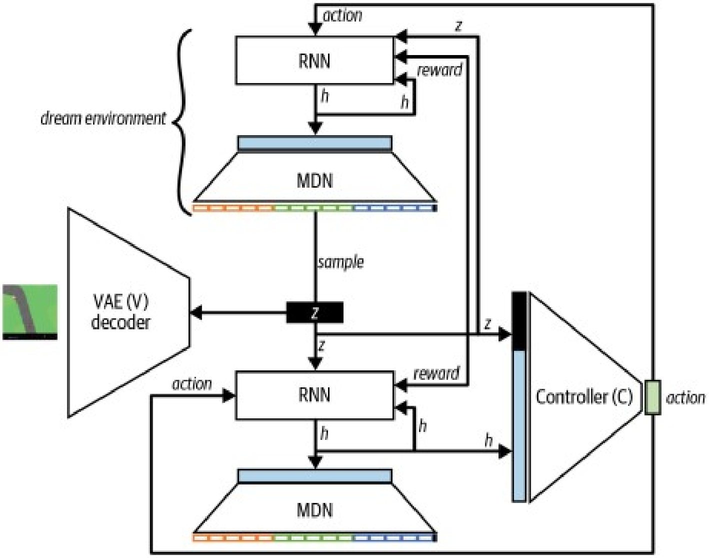

Figure: Training the controller in the MDN-RNN dream environment

TEMPERATURE τ	VIRTUAL SCORE	ACTUAL SCORE
0.10	2086 ± 140	193 ± 58
0.50	2060 ± 277	196 ± 50
1.00	1145 ± 690	868 ± 511
1.15	918 ± 546	1092 ± 556
1.30	732 ± 269	753 ± 139
RANDOM POLICY	N/A	210 ± 108
GYM LEADER	N/A	820 ± 58

Figure: Using temperature to control dream environment volatility